

Classification of central graphical zonotopal algebras

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Abstract

Let G be a finite connected loopless multigraph, and let $A(G)$ be its central graphical zonotopal algebra over the real numbers. We prove that $A(G)$ determines, and is determined by, the graphical matroid obtained by deleting all bridges of G . The algebra isomorphisms need not preserve the grading. The proof extracts the cardinality-weighted cyclic-flat subspaces from the nilpotence lengths of linear elements. It then reconstructs the matroid by rank induction, using intrinsic separation tests, marked virtual edges, and the reduced SPQR decomposition. Parallel multiplicities and undecorated edges of a circuit skeleton are retained throughout the reconstruction. The result addresses the graphical central classification conjecture of Nenashev; no classification of arbitrary real vector configurations or internal zonotopal algebras is asserted.

Keywords: central zonotopal algebra; graphical matroid; cyclic flat; power ideal; SPQR decomposition; matroid reconstruction.

1. Introduction and statement

For a graph G with vertex set $V(G)$ and edge multiset $E(G)$, write $\delta_G(U)$ for the edges with exactly one endpoint in U . Throughout, graphs are finite and have no loops; parallel edges are allowed and counted with multiplicity. Define

$$R_G = \mathbb{R}[x_v : v \in V(G)] / \left(\sum_v x_v \right), \quad A(G) = R_G / I_G, \quad (1.1)$$

where I_G is generated by

$$x_U^{|\delta_G(U)|}, \quad x_U = \sum_{u \in U} x_u, \quad \emptyset \neq U \subsetneq V(G). \quad (1.2)$$

The one-vertex graph has algebra $\mathbb{R}$. All algebra isomorphisms in this paper are unital $\mathbb{R}$ -algebra isomorphisms. Let B_G be the set of bridges, and let $M(G)$ denote the cycle matroid, whose circuits are the edge sets of cycles, including two-edge cycles formed by parallel edges. Isolated vertices do not affect this matroid.

Theorem 1.1. For finite connected loopless multigraphs G and H , the following are equivalent:

1. $A(G)$ and $A(H)$ are isomorphic as algebras;
2. $A(G)$ and $A(H)$ are isomorphic as graded algebras;
3. $M(G \setminus B_G)$ and $M(H \setminus B_H)$ are isomorphic.

In particular, the statement includes trees, bridges, arbitrary parallel classes, and graphs whose bridge-free matroids are disconnected. It does not assert that G itself is determined up to graph isomorphism.

Nenashev classified external graphical zonotopal algebras and formulated the central classification as Conjecture 8 of [N19]. The earlier presentation [N17, §5.1] supplies the forward matroid-to-algebra implication. Conjecture 16 of [N19] concerns arbitrary real matrices and is strictly outside our statement. The central inverse-system description of Holtz and Ron [HR] is a classical input; it should not be confused with a cocircuit-product ideal having related numerical invariants.

Our reconstruction uses subspaces in the dual of the degree-one algebra, not an initially labeled matroid ground set. This distinction matters: intersections of cyclic-flat subspaces can contain directions that are not edges. Moreover, some genuine edge directions move in continuous automorphism orbits when the graph has a two-vertex separation. We therefore recover a matroid decomposition and its gluing data rather than insist on a canonical finite set of all edge directions.

The argument below is self-contained apart from the central inverse-system theorem, the classical forward implication just cited, the vertex form of Menger’s theorem, and the standard reduced SPQR decomposition. We give the representation-uniqueness argument needed in the three-connected step. The SPQR convention is stated explicitly in §7.

1.1. Relation to nearby work and scope of the literature check

The bizonotopal algebras of [KNSV] are different algebras, defined using a doubled edge set. Their graph-reconstruction results are not statements about (1.1). The orbit-harmonics realization in [RRT] and the geometric results of [CL] start from a given graph or arrangement; the statements checked there do not provide the inverse classification in Theorem 1.1.

The source check for this draft used the published statement in [N19], the versions [N17], [KNSV], and [CL] listed below, and their relevant definitions and main results. No direct complete classification covering Theorem 1.1 was located in those sources. The author’s 2018 dissertation [NT] was located but its complete relevant chapter could not be inspected. We consequently make no claim that all related literature has been audited, or that the individual auxiliary constructions below have established priority. This draft is submitted for mathematical scrutiny, not presented as peer-reviewed work.

Earlier internal investigations of low-rank, series-class, and regular three-connected cases are subsumed by the present classification and are not counted as separate results here. The proof does not depend on those internal notes.

2. Algebraic reductions and nilpotence length

A finite-dimensional standard graded algebra A with $A_0 = \mathbb{R}$ is local: its positive-degree ideal $\mathfrak{m}$ is nilpotent, and every element with nonzero constant term is invertible. Since A is generated in degree one,

$$\mathrm{gr}_{\mathfrak{m}} A \cong A. \quad (2.1)$$

Thus an ungraded isomorphism induces a graded one. It suffices to prove the theorem for graded isomorphisms.

Contracting all bridges gives a connected loopless bridgeless graph G_0 , possibly a single vertex, with

$$M(G_0) \cong M(G \setminus B_G). \quad (2.2)$$

By the classical forward implication [N17, Proposition 5.2], $A(G_0) \cong A(G)$. This use of the forward implication does not assume the inverse classification. Henceforth, until §8, graphs are bridgeless. Put

$$n = |V(G)|, \quad m = |E(G)|, \quad r = n - 1, \quad g = m - r.$$

There are no linear generators in (1.2), so the degree-one space $V = A(G)_1$ is the space of vertex potentials modulo constants. Its dual is

$$W = \{(b_v) \in \mathbb{R}^{V(G)} : \sum_v b_v = 0\}.$$

An oriented edge uv represents the vector $b_{uv} = e_u - e_v$ in W . Orientation and nonzero rescaling do not affect its line. For a set of edges F , write $W_F = \text{span}\{b_e : e \in F\}$ and $r(F) = \dim W_F$.

For $a \in V$, define $\ell(a)$ to be the largest j for which $a^j \neq 0$, with $\ell(0) = 0$. Let $Z(a)$ consist of the edges on which the endpoint potentials agree, and count isolated vertices in $c(Z(a))$.

Lemma 2.1 (length formula). The largest nonzero degree of $A(G)$ is g , and

$$\ell(a) = |E \setminus Z(a)| - c(Z(a)) + 1 = g - (|Z(a)| - r(Z(a))). \quad (2.3)$$

Proof. The central inverse-system theorem [HR, Theorem 3.8(5)] identifies the inverse system of I_G with the span of

$$p_Y = \prod_{uv \in Y} (t_u - t_v), \quad E \setminus Y \text{ connected}.$$

The homogeneous differential pairing is nondegenerate, and for $|Y| = j$,

$$\langle a^j, p_Y \rangle = j! \prod_{uv \in Y} (a_u - a_v).$$

Consequently $a^j \neq 0$ precisely when there is a j -element subset $Y \subseteq E \setminus Z(a)$ whose deletion leaves a connected graph. After contracting the components of $Z(a)$, exactly $c(Z(a)) - 1$ edges are needed for a spanning tree. Deleting all other edges of $E \setminus Z(a)$ proves (2.3). A generic potential has $Z(a) = \emptyset$ and length $m - n + 1$; no product in the inverse system has larger degree. The zero potential satisfies the same formula. $\square$

Write $d(a) = g - \ell(a)$ for the defect. Both ℓ and d are intrinsic functions of the graded algebra. The graph interpretation in (2.3) proves their properties but is not input to their extraction.

3. The weighted cyclic-flat data

A flat of a represented matroid consists of all represented elements lying in its span. A flat is cyclic if its restriction has no coloops, equivalently if every one of its edges belongs to a cycle inside the flat. The empty flat is cyclic. Define

$$\mathcal{D}(G) = \{(W_F, |F|) : F \text{ is a cyclic flat of } M(G)\}. \quad (3.1)$$

This is a collection of actual subspaces of W with integer weights, considered up to linear equivalence. It is not a collection of labeled subsets of E .

Proposition 3.1. The algebra $A(G)$ determines $\mathcal{D}(G)$ up to linear equivalence. Conversely, $\mathcal{D}(G)$ determines $A(G)$ up to graded isomorphism.

Proof. For $1 \leq k \leq g$, consider the intrinsic algebraic set

$$D_k = \{a : a^{g-k+1} = 0\} = \{a : d(a) \geq k\}.$$

The zero-edge set $Z(a)$ is a flat. Deleting its bridges leaves a cyclic flat with the same nullity: any edge between vertices of a remaining connected component already lies in $Z(a)$ and cannot be one of its deleted bridges. Therefore

$$D_k = \bigcup_{F \text{ cyclic, } |F| - r(F) \geq k} W_F^\perp. \quad (3.2)$$

Each nonempty cyclic flat F appears as an irreducible component in the layer $k = |F| - r(F)$. Indeed, deleting any element of F lowers nullity by one, and subsequent deletions cannot increase nullity. No proper subset of F has nullity k . A subspace containing $W_F^\perp$ in the union (3.2) would correspond to a flat contained in F , so it must equal F . Conversely every component of a finite union of real linear subspaces is one of its maximal constituent subspaces. Taking annihilators over all layers, removing duplicates, and adding the zero subspace recovers exactly the cyclic-flat subspaces.

For $W_F \neq W$, a generic point $a \in W_F^\perp$ annihilates no edge outside F , and hence

$$|F| = \dim W_F + d(a). \quad (3.3)$$

For $W_F = W$ use $a = 0$; the empty flat has weight zero. This recovers every weight. The rank-zero case has only the zero subspace with weight zero.

For the converse, the cyclic core of $Z(a)$ has its nullity, so

$$d(a) = \max_{W_F \subseteq \ker a} (|F| - \dim W_F), \quad (3.4)$$

where the empty flat is included. Also $m = w(W)$ and $g = m - \dim W$. Thus the data determine the length function. The ideal generated by all pure powers $a^{\ell(a)+1}$ equals I_G : each such power lies in I_G , and each generator a^q from (1.2) is a multiple of the corresponding least vanishing power. A linear equivalence of the data therefore induces an isomorphism of the ideals and quotient algebras. $\square$

In a simple graph, the minimal nonzero cyclic-flat subspaces are exactly the spans of induced cycles. For a connected vertex set S , its edge span is the zero-sum space supported on S , denoted $W(S)$. In particular,

$$\bigcap_i W(S_i) = W\left(\bigcap_i S_i\right), \quad (3.5)$$

with the right side zero when the intersection has at most one vertex. Every cyclic flat is the closure of a union of cycles, so all its subspaces can also be generated as sums of cycle spans. This does not identify the individual edges.

3.1. Restrictions, connectedness, and multiplicity

If F is a cyclic flat, the data of $M|_F$ are exactly the members of $\mathcal{D}(G)$ contained in W_F , with their original weights. Closure agrees because F is a flat.

Lemma 3.2. A bridgeless loopless graphical matroid is disconnected if and only if its data contain two nonzero subspaces U, V such that

$$W = U \oplus V, \quad w(U) + w(V) = m. \quad (3.6)$$

Proof. A partition of the matroid components supplies such cyclic flats. Conversely their edge sets are disjoint because the spans meet only in zero. The weight equality forces them to cover all edges, and the direct sum of spans prevents any circuit from crossing the partition. $\square$

The same test applies inside any restriction. A connected cyclic restriction in a simple graph is, on its nonisolated support, an induced two-vertex-connected graph. This convention about nonisolated support will be used without identifying isolated vertices with matroid elements.

The rank-one nonzero cyclic flats are exactly the parallel classes of multiplicity $\mu \geq 2$. Thus their lines L and multiplicities are known. In a connected matroid of rank at least two, simplification leaves a simple two-vertex-connected graph. Removing all but one copy on every parallel line changes the defect to

$$\bar{d}(a) = d(a) - \sum_L (\mu_L - 1) \mathbf{1}_{a(L)=0}. \quad (3.7)$$

Rank is unchanged; the top degree decreases by $\sum_L (\mu_L - 1)$. Proposition 3.1 recovers the simplified data $\bar{\mathcal{D}}$. We retain the original marked lines and multiplicities; an abstract isomorphism of the simplified matroids alone would not suffice to restore them.

4. The simple three-connected case against arbitrary comparisons

For a simple graph, let $\mathcal{R}_G$ consist of all one-dimensional intersections of nonempty families of induced-cycle spans. This is a finite configuration defined by $\mathcal{D}(G)$. By (3.5), every member of $\mathcal{R}_G$ is the line of $e_u - e_v$ for two vertices, whether or not uv is an edge.

Lemma 4.1. If G is simple and three-vertex-connected, every edge line belongs to $\mathcal{R}_G$.

Proof. For an edge uv , the graph $G - uv$ has two internally vertex-disjoint $u-v$ paths. Otherwise the vertex form of Menger's theorem gives a separating vertex z , and uv is a bridge of $G - z$, contrary to its two-connectivity. Shorten the two paths separately to induced paths. Adding uv yields two induced cycles whose vertex sets meet in exactly $\{u, v\}$. Their spans intersect in the edge line. $\square$

Lemma 4.2 (root-direction comparison). Suppose G is simple and three-vertex-connected. Suppose H' is a simple bridgeless connected graph on the same vertex set, with the same number of edges and the same length function on the vertex-potential space. Then $H' = G$.

Proof. Let $c_v = c(H' - v)$. Formula (2.3) at the single-vertex potential x_v gives

$$\deg_{H'}(v) - c_v = \deg_G(v) - 1. \quad (4.1)$$

Every $c_v \geq 1$. Summing degrees and using equal edge counts forces $c_v = 1$ and equality of degrees for every vertex. For distinct u, v , use $a = x_u + 2x_v$. The three potential values are distinct; putting $c_{uv} = c(H' - \{u, v\}) \geq 1$, the length equality becomes

$$\mathbf{1}_{uv \in E(H')} = \mathbf{1}_{uv \in E(G)} + 1 - c_{uv} \leq \mathbf{1}_{uv \in E(G)}. \quad (4.2)$$

Thus $E(H') \subseteq E(G)$, and equal edge counts give equality. $\square$

Lemma 4.3 (regular representation uniqueness, needed form). Two labeled full-row-rank real representations of the same regular matroid differ by an invertible row transformation and nonzero individual column scalings. No connectivity hypothesis is required.

Proof. Compare any representation with a totally unimodular one. Choose a common labeled basis and put them into the forms $[I \mid D']$ and $[I \mid D]$. The second can be chosen totally unimodular after this normalization. Basis exchanges imply the same zero pattern in D and D' . In the bipartite support graph, an induced cycle gives a square submatrix with exactly two nonzero determinant terms. In D these terms are ± 1 , and total unimodularity forces their sum to vanish. The corresponding determinant for D' vanishes as well, because the two representations have the same bases. Hence the ratios D'_{ij}/D_{ij} have alternating product one around every induced support cycle.

Splitting cycles along chords gives the same condition around all support cycles. Choosing factors along a spanning forest now yields nonzero row and column factors with $D' = RDC$. The assertion follows, with basis-column scalings compensating for the row factors. Disconnected support graphs are handled componentwise. Comparing two representations through the same totally unimodular representation completes the proof. $\square$

The classical uniqueness property is also recalled in [W95, p.223]. The elementary argument is included to specify the exact real, labeled, not-necessarily-three-connected version used here.

Proposition 4.4 (class-external recovery). Let G be simple and three-vertex-connected, and let H be any connected bridgeless loopless multigraph. Every linear equivalence $\mathcal{D}(H) \rightarrow \mathcal{D}(G)$ sends the individual edge lines of H bijectively to those of G . In particular the graphical matroids are isomorphic.

Proof. Since G has no nonzero rank-one cyclic flat, neither does H , which is therefore simple. The ranks and full weights give the same n and m .

First apply Lemmas 4.1 and 4.2 to a data automorphism of G . It permutes $\mathcal{R}_G$, sends every edge to a root direction, and defines a simple graph G' on the original vertices. Linear equivalence and column rescaling preserve ranks and the zero-edge tests, hence the full length function. The graph G' is connected, bridgeless, and has m edges. Lemma 4.2 says $G' = G$. Every data automorphism therefore permutes the actual edge lines.

An automorphism fixing all these lines acts by a common scalar: on each circuit the unique full-support dependency forces the scalars on its edges to agree, and matroid connectedness propagates this equality. Thus the group of data automorphisms modulo scalar matrices is finite. The same holds for H by conjugacy.

For an edge e of H , put

$$W_e = \bigcap_{F \text{ cyclic flat, } e \in F} W_F. \quad (4.3)$$

If $\dim W_e \geq 2$, move the vector of e to a generic vector $b'_e \in W_e$, fixing the other columns. Every old circuit containing e remains dependent, because its closure is cyclic and its other columns already span that closure. Every old independent set remains independent after avoiding finitely many proper subspaces of W_e ; the old b_e witnesses that each avoidance subspace is proper. Thus the labeled matroid is unchanged.

For a cyclic flat containing e , its other columns already span it, so its subspace and weight are unchanged. All other cyclic flats are unchanged as well. Lemma 4.3 gives a row transformation sending every original column line to the corresponding new one. It preserves the data and moves the line of e . Infinitely many choices of the new line give infinitely many projective data automorphisms, a contradiction. Therefore $\dim W_e = 1$ for every edge.

Inside any cyclic flat containing e , shorten a cycle through e along chords while keeping e . The result is an induced cycle through e , also induced in the whole graph because the flat is closed. Hence the intersection in (4.3) equals the intersection of the induced-cycle spans through e . Every edge of H consequently belongs to $\mathcal{R}_H$.

Under the given data equivalence, all these edges become root directions in G 's coordinates. They define a simple connected bridgeless graph H' on $V(G)$, with the same edge count and length function as G . Lemma 4.2 gives $H' = G$. This conclusion concerns the given linear map, not merely the existence of an unmarked matroid isomorphism. $\square$

5. Intrinsic interfaces and marked sides

In §§5–6 the data initially come from a simple bridgeless graph. Connectedness of a cyclic restriction always means matroid connectedness, as tested in Lemma 3.2. Write $w(U)$ for its weight. A minimal cycle space means a minimal nonzero cyclic-flat subspace, hence the span of an induced cycle.

5.1. Two sides without a common edge

Lemma 5.1. Suppose two proper cyclic-flat subspaces U, V satisfy

$$U + V = W, \quad L = U \cap V \text{ is a line,} \quad w(U) + w(V) = m, \quad (5.1)$$

their restrictions are connected, and every minimal cycle space Q lies in U , lies in V , or contains L . Then in every graphical realization the sides are induced two-connected subgraphs, their vertex sets meet in exactly two nonadjacent vertices u, v , and their disjoint edge sets cover the graph. Conversely such a two-side separation satisfies these conditions.

Proof. Let the side supports be A, B . Connectedness of the restrictions identifies their spaces with $W(A), W(B)$. Equations (5.1) force $A \cup B = V(G)$ and $A \cap B = \{u, v\}$. If a is the number of common edges and b the number of edges outside both sides, then $a = b$ and simplicity gives $a \leq 1$.

If $a = b = 1$, the common edge is uv and the outside edge xy joins $A \setminus B$ to $B \setminus A$. The two side graphs remain connected after deleting v . Paths from x and y to u , together with xy , give a cycle avoiding v . Shortening along chords while retaining xy produces an induced cycle still avoiding v . Its span is in neither side and does not contain $L = \mathbb{R}(e_u - e_v)$, a contradiction. Thus $a = b = 0$. The converse follows because every cross-side induced cycle contains both ports. $\square$

5.2. Recovering an added port edge

For the separation of Lemma 5.1, a cross-side induced cycle consists of one induced u – v path in each side. Every pair of such paths is an induced cycle of the whole graph. Hence the intersections $Q \cap U$, as Q ranges over cross-side minimal cycle spaces, are exactly the spans of all induced port paths of the U side.

More generally, suppose a side U and a port line $L \subset U$ have been identified, and these path spans are available. Add one new edge on L , allowing a parallel edge if a port edge already exists. Its entire cyclic-flat space family is obtained by taking all finite sums of the old cyclic-flat spaces in U and the port-path spans, including the zero sum. Indeed, cycles using the new edge correspond to old port paths; cycles not using it are old cycles. If the ports were already adjacent, the only induced port path is that edge, and the added parallel two-cycle supplies the line L . Cycles form cyclic flats after closure, and every cyclic flat is generated by cycles.

For each generated space X , all old vectors in X form a flat Z_X which spans X : every generator of X was already spanned by old edges. Its nullity is that of its cyclic core. Therefore the new weight is

$$w^+(X) = \dim X + \max_{W_F \subseteq X} (w(W_F) - \dim W_F) + \mathbf{1}_{L \subseteq X}. \quad (5.2)$$

This recovers the complete data of the augmented side and its marked port line. The maximum includes the empty flat and uses the old restriction data; it is not an assumption that dimension counts edges.

5.3. Keeping a mark during rank induction

Suppose the classification is known in ranks smaller than r . For two augmented side data of rank less than r related by a linear equivalence preserving the port line L , add K parallel copies to that line in both realizations, with K greater than the number of other edges in either side. As the line already contains an edge, the defect changes by

$$d_{+K}(a) = d(a) + K\mathbf{1}_{a(L)=0}. \quad (5.3)$$

Rank is unchanged. Proposition 3.1 and the induction hypothesis give a matroid isomorphism of the augmented configurations. The marked parallel class is uniquely largest and must be preserved. Deleting the extra copies yields a side isomorphism preserving the marked class; any chosen member can serve as the virtual edge, since parallel elements are interchangeable.

The whole graph in Lemma 5.1 is recovered by the matroid two-sum of the two augmented sides along their virtual edges. Concretely, the circuits are side circuits avoiding the mark and unions of two marked side circuits after removing both marks. Mark-preserving side isomorphisms therefore suffice. Both sides have strictly smaller rank, whereas their edge counts may increase; rank, not edge count, is the induction parameter.

5.4. A common real edge

Lemma 5.2. If connected proper cyclic-flat subspaces satisfy $U + V = W$, $\dim(U \cap V) = 1$, and $w(U) + w(V) = m + 1$, their realizations have exactly one common edge and no outside edges.

Proof. As in Lemma 5.1, the side supports meet in two vertices. With a, b denoting common and outside edges, $a - b = 1$ and $a \leq 1$, whence $a = 1, b = 0$. $\square$

The common edge lies on the known line $L = U \cap V$. Apply the marking procedure to both lower-rank side restrictions and then form their parallel connection, retaining one copy of the common marked edge. Its circuit description uses the side circuits and unions of marked circuits with the common mark removed, so the marked matroids determine the result. For a common parallel class of size μ , first reduce that class to one edge on each side, leave all other multiplicities unchanged, glue, and restore μ copies once.

6. Circuit skeletons, bare edges, and parallel layers

6.1. A circuit skeleton with every position decorated

Lemma 6.1. Let $U_1, \dots, U_k$ be nonzero cyclic-flat subspaces with connected restrictions, where $k \geq 3$. Suppose the addition map

$$\bigoplus_{i=1}^k U_i \longrightarrow W \quad (6.1)$$

is onto, has a one-dimensional kernel whose projection to every U_i is nonzero, and $\sum_i w(U_i) = m$. Then every realization consists of the side graphs arranged on a circuit skeleton, with two distinct ports on each side and no other edges. The kernel projects onto its port line L_i in each side.

Proof. Pairwise intersections of the U_i vanish, since a nonzero intersection would give a kernel vector supported on two positions. Their edge sets are thus disjoint and, by weight, cover all edges. Let A_i be the support of side i . Two supports meet in at most one vertex.

Form the bipartite incidence graph with side nodes i and shared-vertex nodes z , joining i to z when $z \in A_i$. It is connected. If $n = |\bigcup_i A_i|$, its cycle rank is

$$\sum_i (|A_i| - 1) - (n - 1) = 1. \quad (6.2)$$

A relation $\sum_i u_i = 0$ vanishes along every tree branch of this incidence graph. At a leaf side its private vertex coordinates vanish; its one remaining shared coordinate must also vanish because the coordinate sum is zero. Peeling the branch proves the assertion. Full support of the kernel forces every side onto the unique cycle. Shared-vertex nodes have degree at least two, and there can be no branch without a side node. Thus the entire incidence graph is an alternating cycle. Its relation on side i is the difference of its two port coordinates. $\square$

Cross-side induced cycles choose one induced port path independently in every side. If the ports are adjacent, that edge is the unique induced port path. Consequently their intersections with U_i give all port-path spans, and (5.2) recovers each augmented side, including existing port-edge multiplicity. Every side has rank smaller than $\dim W$. Using §5.3, the matroid is the two-sum of these marked sides with a k -element circuit skeleton. The circuit matroid permits any permutation of its elements, so the geometric order of side positions is unnecessary.

6.2. A circuit skeleton with bare edges

Assume now that the whole simple graphical matroid is connected. Suppose connected nonzero cyclic restrictions have mutually direct spans $U_1, \dots, U_k$, with $k \geq 1$, and put

$$b = m - \sum_i w(U_i) \geq 1, \quad \dim W - \sum_i \dim U_i = b - 1. \quad (6.3)$$

Lemma 6.2. These conditions force a circuit skeleton consisting of the k sides and b additional bare edges. Every side has two distinct ports; no side is attached off the circuit.

Proof. The side edge sets are disjoint. Their union has precisely these matroid components and thus a forest of graph blocks. Contract a spanning tree in each side and discard loops arising from its internal cycles. The b remaining edges give a connected graph of rank $b - 1$. They have no bridges: contraction cannot make a surviving non-coloop a coloop, since dually deletion cannot create a loop. Hence they form a circuit, permitting a single loop when $b = 1$ or two parallel edges when $b = 2$.

Each contracted vertex came from a component of the side-block forest, possibly an isolated original vertex. The incoming and outgoing circuit edges are connected inside it by a unique block path. A side outside that path attaches to the rest through one cut vertex and would be a separate matroid component, contradicting connectedness of the whole matroid. The two ports in each side on the path are distinct for the same reason. Expanding gives the claimed skeleton. $\square$

It remains to find the ports intrinsically. Let $\mathcal{Q}$ be the minimal cycle spaces not contained in any side and put $P = \bigcap_{Q \in \mathcal{Q}} Q$. Cross-side induced cycles consist of all bare edges and an independently chosen induced port path in every side. Such cycles exist. Write N for the span of the bare edges. Since at least one side is present, the bare edges alone are independent, so $\dim N = b$. The subspace $N \cap \bigoplus U_i$ is one-dimensional, with nonzero projections to all port lines.

In a two-connected side, the intersection of all induced port-path vertex sets is exactly its two ports. For nonadjacent ports this follows from two internally disjoint paths and shortening; for adjacent ports use the edge itself. Thus the intersection of path spans is L_i . Quotienting by $N + \sum_i L_i$, the images of U_i/L_i are in direct sum. Independence of the path choices gives

$$P = N + \sum_i L_i, \quad \dim P = b + k - 1, \quad P \cap U_i = L_i. \quad (6.4)$$

Moreover $Q \cap U_i$ is the span of that cycle's port path, and all such spans occur. Formula (5.2) supplies the augmented side data. Each side has strictly smaller rank: the only possible exception is $k = b = 1$, which would make a proper flat span all of W , an impossibility. The whole matroid is reconstructed from the marked sides and a $(k + b)$ -element circuit by two-sums at the k marked positions. Its b bare positions may be permuted arbitrarily. The case with no sides is simply a circuit and is treated as a base case.

6.3. Applying the interfaces to multigraphs

Use the simplified data $\overline{D}$ from (3.7) to locate the interfaces of §§5–6. A side cyclic flat in the simplification corresponds to a cyclic flat with the same span in the original multigraph. Its original restriction data retain all multiplicities. Generate augmented-side spaces using the port paths extracted from $\overline{D}$ and all original restriction cyclic flats; compute their weights by (5.2) using the original weights. This yields the original multiple-edge side with one new virtual edge, rather than an unmarked simplified substitute.

For a bare skeleton, every parallel class on a bare edge is among the known rank-one cyclic flats of the original data. Its line lies in P and is not contained in any U_i . Conversely a known parallel class not in a side belongs to a bare position, because the skeleton exhausts the edges. Record their multiplicities as a multiset and fill the remaining b positions with ones. Circuit symmetry makes the multiset sufficient; the side multiplicities are retained in the full side data. If the simplification is itself a circuit, the multiset of all edge multiplicities similarly determines the original matroid.

In a common-real-edge interface use the reduction and restoration described after Lemma 5.2, preventing double counting of the shared parallel class. These operations introduce no call to the classification at the same rank.

7. Exhaustiveness of the interfaces

We use the reduced SPQR convention in [HIKLR, Definition 3, pp.46:5–46:6]. For a two-connected multigraph with at least three edges, it is a tree of skeletons of the following types: S , a simple cycle of length at least three; P , two vertices with at least three parallel edges; and R , a simple three-connected graph. A tree edge corresponds to one virtual edge in each adjacent skeleton. Original edges appear in exactly one skeleton; we do not add separate Q leaves. There are no adjacent S nodes and no adjacent P nodes. Gluing corresponding virtual edges at their endpoints and deleting them recovers the graph. Each expansion network together with its port edge is two-connected. The classical source is [DT]; the convention quoted in [HIKLR] applies to general two-connected graphs, although that paper's later dynamic algorithms concern planar graphs.

Lemma 7.1 (coverage). Every simple two-connected graph is a circuit, is three-connected, or admits at least one interface of Lemma 5.1, Lemma 5.2, Lemma 6.1, or Lemma 6.2.

Proof. First consider an expansion incident to an R or P skeleton, with the parent virtual edge removed. An R skeleton remains two-connected after deleting an edge. A P skeleton then has at least two branches. Replacing an edge of a two-connected skeleton by a network which becomes two-connected after adding its port edge preserves two-connectivity: after deleting an internal

vertex every residual network component reaches a port, and the rest of the skeleton supplies the alternate port connection. Deleting a port leaves the network connected to its other port. The same argument treats two or more parallel branches. Thus the full expansion on an R or P side, without its parent edge, is two-connected.

If the reduced tree has an S node, let its virtual edges expand to k side networks and let its remaining b edges be real, so $k + b \geq 3$. No adjacent node is S ; hence all expanded sides are two-connected by the preceding paragraph. Each is induced on its vertex support: all edges at interior vertices belong to that expansion, and the other part of the skeleton supplies no additional edge on the same two ports. Their edge sets are connected cyclic flats.

If $k = 0$, the whole graph is a circuit. If $b = 0$, the side spaces have a one-dimensional full-support sum relation and their weights sum to m , as required in Lemma 6.1. If $b \geq 1$ and $k \geq 1$, the bare edges break the relation among side spaces: their sum is direct, the rank deficit is $b - 1$, and the weight deficit is b . These are (6.3).

If there is no S node, a single R node is the three-connected case. A single P node with only real edges cannot realize a simple graph. Otherwise take a tree edge, whose two adjacent skeletons are P or R . Their expansions after deleting the virtual edge are two-connected, share exactly the two ports, and have disjoint interiors. Each is induced except possibly for a real edge between the ports assigned to the other side.

If no real port edge exists, the two sides give Lemma 5.1. If one exists, add this same edge to the other side as well. Both sides are now induced, cover the graph, and share exactly that edge, giving Lemma 5.2. Every side contains an interior vertex: a simple graph cannot supply two real parallel branches on a P side. Therefore both side ranks are strictly smaller. This proves coverage. $\square$

The role of SPQR is solely to prove that a valid interface exists in one realization. The algorithmic tests in §§5–6 use the data itself. No alignment of two input SPQR trees is assumed.

8. Proof of Theorem 1.1

By §2, it suffices to compare bridgeless connected loopless graphs with linearly equivalent data. We prove their matroids isomorphic by strong induction on $r = \dim W$, allowing arbitrary parallel multiplicity at each rank.

At rank zero the matroid is empty. At rank one a connected bridgeless loopless matroid is one parallel class, and the full weight determines its size. If the data admit (3.6), the same subspace pair partitions both realizations into direct summands of smaller rank. Their restriction data agree. A restriction graph need not be connected; discard isolated vertices and identify one chosen vertex from each nonempty component into one common vertex. This connected loopless realization preserves the matroid direct sum, rank, and data. The induction hypothesis therefore applies to each restriction within the stated connected input domain. Combining the resulting isomorphisms treats this case.

Otherwise both matroids are connected. Assume $r \geq 2$ and simplify using (3.7), retaining the original weighted lines. Both simplifications are simple two-connected graphs. Choose one and apply Lemma 7.1.

If it is a circuit, its simplified nullity is one. The comparison simplification has the same nullity and a connected matroid, so it too is a circuit. The original multiplicity multiset determines the parallel-expanded circuit in each realization, and the two multisets agree.

If it is three-connected, Proposition 4.4 applies to the given linear equivalence of simplified data. That equivalence matches the actual edge lines. All marked parallel lines and their original weights were retained in (3.7), so the correspondence restores each original parallel class with its multiplicity.

This gives the required isomorphism, not merely an arbitrary isomorphism of simplifications.

In every other case choose the data-defined interface supplied by Lemma 7.1. The intrinsic recognition lemmas force the same type of interface in the comparison realization. Section 6.3 reconstructs the full multiple-edge side data and any bare multiplicity multiset. All augmented sides have strictly smaller rank. Apply §5.3 and the induction hypothesis after making the marked parallel class uniquely largest. This obtains mark-preserving side matroid isomorphisms.

For Lemma 5.1 glue by a two-sum; for Lemma 5.2 use the parallel connection and restore the shared multiplicity once; for Lemmas 6.1–6.2 attach the marked sides to the common circuit skeleton and restore the bare multiplicities. The circuit descriptions in §§5–6 show that the resulting matroids are isomorphic. Extra parallel copies used to mark ports increased only the edge count, never the rank. Thus no same-rank invocation has entered the induction.

This proves the inverse implication for all bridgeless inputs. Equation (2.2) restores bridges. The forward implication follows from [N17, Proposition 5.2], and (2.1) establishes equivalence of graded and ungraded isomorphism. $\square$

9. Checks, interpretation, and limitations

The proof is not a finite enumeration argument. Two small controls help distinguish its data from weaker substitutes.

First, take triangles with edge multiplicities $(2, 2, 4)$ and $(2, 3, 3)$. Both have rank two, top degree six, and the same unweighted union of three equal-potential lines. Their generic defects on those lines are respectively the multisets $\{1, 1, 3\}$ and $\{1, 2, 2\}$. Thus the full data distinguish them; omitting the weights would incorrectly identify different parallel-class multisets.

Second, in $K_{3,4}$, induced cycles with vertex sets $\{a, b, 1, 2\}$ and $\{a, b, 3, 4\}$ have spans intersecting in $\mathbb{R}(e_a - e_b)$, although ab is not an edge. This is why §4 treats the one-dimensional intersections as candidates and proves an additional comparison lemma, rather than declaring them to be the edge set.

During development, direct rational cut-power calculations checked the first example, and bounded edge-subset calculations checked representative circuit-skeleton, bare-path, and parallel-edge interfaces. Those checks support only their finite inputs. Every general implication used in Theorem 1.1 is proved above or identified as a classical dependency; no enumeration output is needed to apply the theorem.

The result has a natural boundary here: the root-direction argument and the exhaustive graphical decomposition use graphical structure. They do not establish the general real-matrix conjecture of [N19]. Nor have we extended the frozen real field to all characteristic-zero fields. Such extensions would require their own statements and checks.

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AI tools assisted with exploration, drafting, and computational checks. Carptopus is the author and takes responsibility for the claims and their presentation. The manuscript has not undergone external peer review.

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